

From Data to Business Decision

End-to-End Data Analysis Practice Activity

Scenario

You are a Junior Data Analyst for an online retail company. The business wants to understand customer purchasing behaviour and eventually predict whether a customer is likely to make a purchase.

Practice Dataset

Use the following intentionally messy customer dataset. Your task is to work through the complete data-analysis lifecycle. Do not immediately fix everything—first identify what you observe and explain why each step is needed.

Customer_ID	Age	Gender	City	Income	Visits	Avg_Spent	Purchase	Signup_Date
C001	25	Female	Toronto	52000	5	45.5	Yes	2025-01-10
C002	34	Male	Toronto	68000	8	72.3	No	2025-01-15
C003	29	F	Brampton	55000	6	61.2	Yes	2025/01/20
C004	42	Male	Mississauga	82000	12	95.0	Yes	2025-02-01
C005	31	Female	Toronto	NaN	4	38.5	No	2025-02-05
C006	31	Female	Toronto	61000	7	55.75	Yes	2025-02-08
C007	150	Male	Brampton	59000	5	49.0	No	2025-02-10
C008	27	M	Toronto	47000	3	35.25	No	2025-02-12
C009	36	Male	Toronto	71000	10	88.4	Yes	2025-02-15
C010	29	Female	Brampton	54000	6	NaN	Yes	2025-02-18
C011	34	Male	Toronto	68000	8	72.3	No	2025-01-15
C012	24	Female	Toronto	45000	2	29.99	No	2025-02-20

The 8-Part Data Journey

Stage	What You Do	Key Skills / Concepts	Goal
1. Access & Filter	Load, inspect, select and filter the DataFrame.	DataFrame basics; head(), shape, columns, info(); loc; iloc; Boolean filtering; isin(); between().	Identify relevant records based on a business question.
2. Data Wrangling	Identify and address data-quality issues.	Missing values; duplicates; inconsistent categories; incorrect data types; dates; invalid values; outliers.	Make the dataset reliable and analysis-ready.
3. EDA	Explore patterns, distributions and relationships.	describe(); value_counts(); groupby(); aggregations; comparisons.	Generate questions and identify useful patterns.
4. Visualization	Turn findings into visual evidence.	Bar, line, scatter, histogram and box plot; chart selection based on the question and variable types.	Communicate patterns, comparisons, trends and relationships.
5. Feature Engineering	Create useful variables for modeling.	New columns; transformations; encoding; combining existing variables.	Create features that may help explain or predict the target.
6. Modeling	Build a predictive model.	Features (X); target (y); train/test split; model selection; training; prediction.	Use the prepared data to make predictions.
7. Evaluation	Assess model performance.	Accuracy; precision; recall; F1 score; confusion matrix.	Determine whether the model is performing reliably.
8. Recommendation	Translate analysis into business action.	Findings → evidence → recommendation; communicate limitations and implications.	Turn analytical results into a practical business decision.

Part 1 — Accessing & Filtering Data

Goal: Become comfortable navigating and retrieving information from a DataFrame.

- Load the dataset into a DataFrame.
- Inspect the first few rows and overall structure.
- Identify the number of rows and columns.
- Review column names and data types.
- Select individual and multiple columns.
- Practice selecting rows by position and by label/condition.
- Filter records using one condition.
- Filter records using multiple conditions with AND / OR logic.
- Practice filtering using category membership and numeric ranges.

Hint: Start with the question you want to answer, then decide which rows and columns you need.

#	Question	Hint
1	How many rows and columns are in the dataset?	Think shape.
2	Which columns are available, and what data type does each contain?	Use columns and info().
3	Display only Customer_ID, Age, City, and Purchased.	Select multiple columns.
4	Which customers are from Toronto?	Boolean filtering.
5	Which customers visited more than 6 times?	Use a numerical condition.
6	Which customers are from Toronto and purchased?	Think &.
7	Which customers have income above 60,000 or more than 8 visits?	Think `

Business Question:

Which customer information do we need to calculate our purchase-rate KPI and identify potential high-value customers?

Part 2 — Data Wrangling

Goal: Identify and resolve problems that could affect the reliability of the analysis.

- Check for missing values and decide how they should be handled.
- Check for duplicate records and determine whether they represent true duplicates.
- Look for inconsistent category labels and standardize them where appropriate.
- Check whether numerical, categorical and date columns have appropriate data types.
- Convert date fields into a consistent datetime format.
- Investigate suspicious or invalid values.
- Identify potential outliers and investigate their cause before deciding what to do with them.
- Validate the dataset after changes.

Hint: An outlier is not automatically bad data. Ask whether it is an error, a legitimate unusual observation, or something that needs further investigation.

#	Question	Hint
1	Which columns contain missing values, and how many are missing in each?	<code>isnull()</code> / <code>sum()</code> .
2	How would you handle the missing Income value? Explain your choice.	Don't automatically delete it.
3	How would you standardize the Gender values so that F, female, and Female are consistent?	Think mapping/replacement.
4	Are there duplicate customer records? Identify and handle them appropriately.	Check <code>duplicated()</code> .
5	Which columns have incorrect or inappropriate data types?	Inspect <code>dtypes</code> .
6	Convert <code>Signup_Date</code> into a proper datetime format.	Think <code>pd.to_datetime()</code> .
7	Customer C007 has an age of 150. What would you do with this value?	Investigate before removing. Is every outlier necessarily bad?

Business Question:

What data-quality issues could make our KPI or customer-targeting analysis unreliable?

Part 3 — Exploratory Data Analysis (EDA)

Goal: Understand what the cleaned data is telling you before building a model.

- Review descriptive statistics for important numerical variables.
- Examine category frequencies.
- Compare groups using appropriate aggregations.
- Look for differences between customers who purchased and those who did not.
- Investigate relationships between numerical variables.
- Look for unusual patterns, distributions and potential relationships.
- Record observations that could lead to further investigation.

Hint: EDA is about asking questions and discovering patterns—not proving that one variable causes another.

#	Question	Hint
1	What are the mean, minimum, maximum and median values for Age, Income and Avg_Spend?	describe().
2	How many customers purchased versus did not purchase?	value_counts().
3	What is the average number of visits for customers who purchased versus those who did not?	groupby().
4	Which city has the highest number of customers?	Count by City.
5	Does average spending differ between purchasers and non-purchasers?	Group by Purchased.
6	Is there an apparent relationship between Income and Avg_Spend?	Compare the two numerical variables.
7	Based on your EDA, identify two patterns that may be useful for the business .	Don't just calculate—interpret.

KPI & Metric Analysis

Calculate:

KPI 1 — Purchase Rate

Purchase Rate =

Number of Customers Who Purchased

Total Number of Customers

Ask:

What is the overall purchase rate?

Then:

Does the purchase rate differ by City?

Does the purchase rate differ by Income group?

Does the purchase rate differ between customers with higher and lower visit frequency?

Metric 2 — Average Visits

Ask:

How does average visit frequency differ between purchasers and non-purchasers?

This helps learners connect:

Visits → customer engagement → potential purchase behavior

Metric 3 — Average Spend

Ask:

Do customers who purchase have a different average spend than customers who do not?

Metric 4 — Customer Value

You already create:

$\text{Customer_Value} = \text{Visits} * \text{Avg_Spend}$

Now connect it explicitly to the business:

Which customers have the highest potential customer value?

And:

Would you prioritize a customer with a high purchase probability or a customer with high potential value? Why?

That is a **much better analyst-level question** than simply asking them to multiply two columns.

Part 4 — Visualization

Goal: Select and create visualizations that clearly communicate EDA findings.

Question / Goal	Possible Visualization	Hint
Compare categories	Bar chart	Use for comparing values across discrete groups.
Show a trend over time	Line chart	Use when order/time is important.
Show relationship between two numerical variables	Scatter plot	Look for patterns, direction, strength and unusual observations.
Show a distribution	Histogram	Look at shape, concentration and spread.
Show spread and potential outliers	Box plot	Compare distributions across groups.
Show proportions	Pie chart (sparingly)	Use only when categories

		represent meaningful parts of one total.
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Hint: Do not choose a chart because it looks attractive. Choose it based on the analytical question and the types of variables involved.

From KPI to Visualization

Business Need	KPI / Metric	Recommended Visualization
Understand overall purchasing	Purchase Rate	KPI card / bar
Compare purchasing by city	Purchase Rate by City	Bar chart
Understand customer engagement	Average Visits by Purchase Status	Bar chart
Understand spending	Avg_Spend by Purchase Status	Box plot
Identify relationship	Income vs Avg_Spend	Scatter plot
Identify high-value customers	Customer Value	Histogram / bar chart
Understand customer distribution	Income	Histogram

Then add:

Your visualization should answer a business question, not simply display a variable.
That's an important professional habit.

#	Question	Hint
1	Which chart would you use to compare the number of customers across cities?	Compare categories.
2	Which chart would best show the distribution of customer income?	Think distribution.
3	Which chart would you use to examine the relationship between Income and Avg_Spend?	Two numerical variables.
4	Which chart would help compare spending across different cities?	Think categories + distribution.
5	Which chart would best show purchasing trends if you had monthly purchase data?	Time → trend.
6	Create one visualization that supports an important finding from your EDA.	Choose based on your finding.

#	Question	Hint
7	Looking at your visualization, what business insight can you communicate?	Chart → observation → business meaning.

Part 5 — Feature Engineering

Goal: Transform existing information into useful features for modeling.

- Identify which variables could be useful predictors.
- Create meaningful calculated features from existing columns.
- Consider whether variables need transformation.
- Convert categorical variables into a form suitable for modeling.
- Consider whether numerical variables require scaling for the selected algorithm.
- Check for potential data leakage.
- Review the resulting features for quality and relevance.

Hint: Feature engineering is different from cleaning. Cleaning fixes data-quality problems; feature engineering creates or transforms variables for analytical or predictive use.

#	Question	Hint
1	What new feature could represent the customer's overall spending value?	Combine Visits and Avg_Spend.
2	Create a Customer_Value feature.	Think multiplication.
3	How could you convert Purchased from Yes/No into a numerical target?	Mapping.
4	Would City and Gender be directly usable by every machine-learning algorithm? If not, what could you do?	Think encoding.
5	Could Income be transformed into meaningful income groups? What groups would you create?	Think business categories.
6	Which existing variables do you think are likely to be useful predictors of purchasing? Why?	Think business relevance.
7	Before using your engineered features for modeling, what	Quality, leakage, missing

#	Question	Hint
	would you check?	values, relevance, etc.

Part 6 — Modeling

Goal: Use prepared features to build a model that addresses the business question.

- Define the target variable.
- Select the features that will be used as inputs.
- Separate features (X) from the target (y).
- Split the data into training and testing sets.
- Choose a model appropriate for the type of target and business problem.
- Train the model using the training data.
- Generate predictions on unseen test data.

Hint: The model should be chosen because it fits the problem—not simply because it is familiar.

#	Question	Hint
1	What is the target variable you are trying to predict?	What represents the business outcome?
2	Which variables would you use as features?	Think potential predictors.
3	Separate your data into X and y. What does each represent?	Features vs target.
4	Why should we split the data into training and testing sets?	Think unseen data.
5	What type of machine-learning problem is this?	Yes/No outcome.
6	Which model could you use to predict the outcome?	Consider classification models.
7	Train the model and generate predictions for the test set.	Think <code>.fit()</code> and <code>.predict()</code> .

Modeling Business Objective

Business Question:

Can we predict which customers are most likely to purchase so that marketing can focus its limited budget on the most promising customers?

Target KPI:

Purchase Rate among targeted customers

Model Output:

Probability that each customer will purchase.

Part 7 — Evaluation

Goal: Determine whether the model performs well enough for its intended use.

Metric / Tool	Question to Consider
Accuracy	How often is the model correct overall?
Precision	When the model predicts the positive class, how often is it correct?
Recall	Of the actual positive cases, how many did the model identify?
F1 Score	How well are precision and recall balanced?
Confusion Matrix	What types of correct and incorrect predictions is the model making?

Hint: Do not rely on accuracy alone. Consider the business impact of false positives and false negatives.

#	Question	Hint
1	What is the model's accuracy?	Use an appropriate metric.
2	How many predictions were correct and incorrect?	Think confusion matrix.
3	What is precision, and what does it tell you about this model?	Focus on predicted positives.
4	What is recall, and what does it tell you?	Focus on actual positives.
5	What is the F1 score?	Balance precision and recall.
6	Would accuracy alone be enough to decide whether this model is useful? Why or why not?	Think class imbalance and business consequences.

#	Question	Hint
7	If the model will be used by the marketing team, which type of error might matter more: false positives or false negatives? Explain.	Connect evaluation to business impact.

Business Perspective: Connect model evaluation to the business decision

Your current evaluation section is technically correct, but you can add this table:

Model Metric	Technical Meaning	Business Meaning
Accuracy	Overall correct predictions	How often is the model right?
Precision	Correctness of positive predictions	If marketing targets someone, how often are they actually likely to purchase?
Recall	Ability to identify actual purchasers	How many potential purchasers are we successfully finding?
F1	Balance between precision and recall	How well does the model balance targeting efficiency and coverage?
Confusion Matrix	Types of prediction errors	What happens when marketing targets the wrong customer or misses a potential buyer?

Then your existing question becomes much stronger:

If the marketing budget is limited, would you prioritize precision or recall? Explain why.

Part 8 — Business Recommendation

Goal: Translate technical analysis into an understandable business recommendation.

- State the key finding.
- Support the finding with data or visualization.
- Explain what the model or analysis suggests.

- Connect the result to the original business question.
- Recommend a practical action.
- State important limitations or assumptions.

Final Challenge

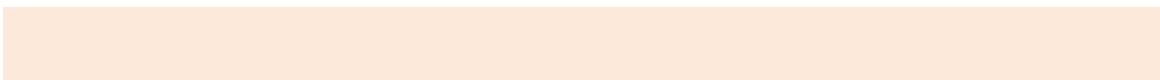
The marketing team has a limited budget and wants to target customers who are most likely to purchase. Use the complete workflow to investigate the data and prepare a short recommendation.

1. Identify the business question and relevant data.
2. Inspect and filter the DataFrame.
3. Wrangle and validate the data.
4. Perform EDA and identify useful patterns.
5. Create visualizations that support your findings.
6. Engineer appropriate features.
7. Build a suitable predictive model.
8. Evaluate the model using appropriate metrics.
9. Communicate a recommendation supported by evidence.

Remember the Framework

ACCESS → WRANGLE → EXPLORE →
VISUALIZE → ENGINEER → MODEL →
EVALUATE → RECOMMEND

Data Wrangling makes the data trustworthy. EDA helps us understand the data. Visualization helps us see and communicate patterns. Feature Engineering prepares useful inputs for modeling. Modeling makes predictions. Evaluation tells us whether those predictions are reliable. Recommendations connect the analysis back to the business.



Additional Information & Resources

Use the information below as additional context when completing the activity. Refer to it when you need clarification about the business context, KPIs, metrics, or analytical decisions.

Business Context: From Business Goal to KPI

Business Goal:

The marketing team has a limited budget and wants to increase purchases by targeting customers who are most likely to convert.

Before analyzing the data, translate the business goal into measurable KPIs and supporting metrics.

Business Goal	KPI	Supporting Metrics	Data Needed	Business Question
Increase purchases	Purchase Rate / Conversion Rate	Number of purchases, number of customers	Purchased, Customer_ID	What proportion of customers purchase?
Identify high-value customers	Customer Value	Avg_Spend, Visits	Visits, Avg_Spend	Which customers generate the most potential value?
Improve	Predicted	Precision, Recall,	Purchased +	Can we identify

Business Goal	KPI	Supporting Metrics	Data Needed	Business Question
targeting	Purchase Rate	F1	customer features	customers likely to purchase?
Increase marketing efficiency	Conversion by Segment	Purchase rate by City, Income group, Gender	City, Income, Gender, Purchased	Which customer segments are more likely to purchase?
Understand customer engagement	Average Visits	Visits by Purchased status	Visits, Purchased	Are more engaged customers more likely to purchase?
Increase revenue opportunity	Potential Customer Value	Visits × Avg_Spend	Visits, Avg_Spend	Which customers represent the greatest potential value?

Add this key distinction

KPI = business outcome we want to improve.

Metric = measurement that helps us understand or influence that outcome.

For example:

Business Goal: Increase purchases

KPI: Purchase Rate

Metrics: Visits, Avg_Spend, Income, City, Customer Value

Analysis: Compare these metrics between purchasers and non-purchasers

Decision: Target customer segments with higher likelihood and/or value

2. Map each of your 8 stages to the KPI

I would modify your **8-Part Data Journey** table like this:

Stage	What You Do	Key Skills / Concepts	Business Connection
1. Access & Filter	Load, inspect, select and filter the DataFrame.	DataFrame basics, filtering, loc, iloc, Boolean logic	Identify the data needed to answer the KPI/business question.
2. Data Wrangling	Identify and address data-	Missing values, duplicates, categories,	Ensure KPI calculations and decisions are based

Stage	What You Do	Key Skills / Concepts	Business Connection
	quality issues.	dates, outliers	on trustworthy data.
3. EDA	Explore patterns, distributions and relationships.	describe(), groupby(), aggregations	Identify factors associated with purchasing and customer value.
4. Visualization	Turn findings into visual evidence.	Bar, line, scatter, histogram, box plot	Communicate KPI performance and segment differences.
5. Feature Engineering	Create useful variables for modeling.	New columns, transformations, encoding	Create variables that may improve prediction and business segmentation.
6. Modeling	Build a predictive model.	X/y, train/test split, classification	Predict customers who are likely to purchase.
7. Evaluation	Assess model performance.	Accuracy, precision, recall, F1, confusion matrix	Determine whether the model is reliable enough to support targeting decisions.
8. Recommendation	Translate analysis into action.	Findings → evidence → recommendation	Decide who to target, why, and what action to take.